SUPPORTING ADULT GUIDANCE

Forming and solving a pair

Equations, Inequalities and Simultaneous Relationships · Lesson 9. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson brings the unit together: **forming** a pair of simultaneous equations from a real two-unknown situation, **solving** it (by the learner's choice of method), and **interpreting** the result in context. It combines the modelling of Lesson 4 with the solving of Lessons 7–8. The contexts are genuine budgeting/shopping problems (Mode A).

Driving Question: *How do I set up two equations from one situation?*

The mathematics, in plain terms

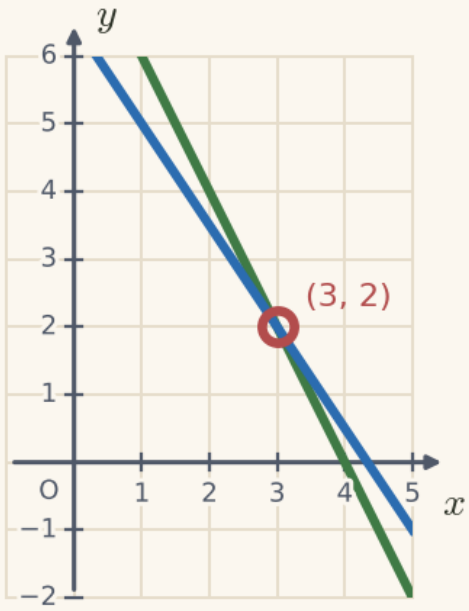
A situation with two unknowns supplies two independent linear relationships; naming the unknowns and translating each relationship gives a solvable pair. Solving (substitution or elimination) yields both values, which are then interpreted in context and checked against both original statements. Because the system is fully determined, the values also price any further combination of the two items.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — A café order

Two lunches and a drink cost £8; three lunches and two drinks cost £13. Let x = lunch price, y = drink price.

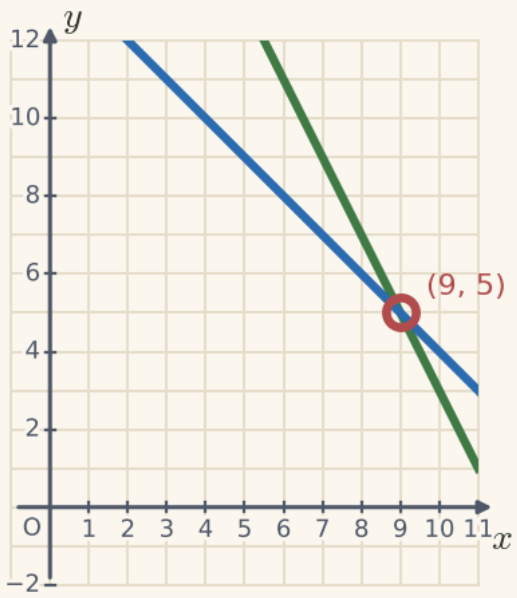


1. Form an equation from each sentence.	$2x + y = 8$ and $3x + 2y = 13$
2. Eliminate y: double the first, then subtract.	$(3x + 2y) - (4x + 2y) = 13 - 16 \rightarrow x = 3$
3. Find y and interpret.	$y = 2$ — a lunch is £3, a drink £2
4. Check both facts.	$2\times3 + 2 = 8$ ✓ and $3\times3 + 2\times2 = 13$ ✓

A lunch is £3 and a drink £2 — formed as $2x + y = 8$ and $3x + 2y = 13$, solved and checked in both facts.

Example 2 — Tickets

Two adult tickets and a child ticket cost £23; one adult and one child cost £14.



1. Form the pair (x = adult, y = child).	$2x + y = 23$ and $x + y = 14$
2. Subtract — the y matches.	$(2x + y) - (x + y) = 23 - 14 \rightarrow x = 9$
3. Find y and interpret.	$y = 5$ — an adult is £9, a child £5
4. Check both facts.	$2\times9 + 5 = 23$ ✓ and $9 + 5 = 14$ ✓

An adult ticket is £9 and a child £5 — subtracting removes y directly; both facts check.

Name, form both, solve, interpret, check

Insist on naming the two unknowns first ('let x = …, y = …'). Then form one equation per fact — the hardest step, so give it time and use the two sentences explicitly. Solving is now routine from Lessons 7–8; let learners choose the method and justify it. Finish by stating each value in context and checking both facts.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Turn each sentence of a situation into an equation — forming the pair.	~12 min

Movement	Solve the formed pair by elimination, then read the values.	~10 min
Reflection	Use the found values to price a new combination.	~9 min
Creativity	Invent a two-item situation from chosen prices.	~8 min
Rest	A short pause on two facts pinning down two unknowns.	~2 min
Nutrition	Contextual (Mode A): real budgeting — pricing items from receipts.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Not naming the unknowns, so the equations are ambiguous.
Insist on 'let x = …, y = …' first. 'What exactly do x and y stand for here?'

Forming only one equation, or reusing the same fact twice.
Two unknowns need two *different* facts. 'You have used the first sentence — what does the second one tell you?'

Mis-reading a coefficient (e.g. 'three lunches' as +3 total, not 3x).
Tie each number to its item: 'Three lunches at £x each is how much — in terms of x?'

Solving but not interpreting or checking both facts.
'What is each item worth, in pounds? And do your prices make both sentences true?'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: writing expressions from words, and solving a pair by substitution or elimination (Lessons 7–8). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner writes one equation, or reads 'three lunches' as a constant. The aim is a question that returns the modelling to them.

- "What do x and y stand for — in words, with units?"
- "The first sentence gives one equation. What different fact does the second give?"
- "Three lunches at £x each — how do you write that? And the total?"

- “What is each item worth, and do your prices make both sentences true?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Which method should they use to solve?

Either. Elimination is quick when an item's coefficient already matches across the two equations (subtract); substitution suits a variable easily isolated. Let learners choose and justify — both give the same prices.

Why must there be two facts?

Two unknowns have two degrees of freedom; one equation leaves a whole line of possibilities, and it takes a second, independent equation to pin down a single pair — the crossing of the two lines.

Do prices have to be whole numbers?

No — the stretch gives £2.50. Money problems often have fractional (pence) answers; the method is unchanged.

What if the two facts don't give a unique answer?

If the two equations describe the same relationship (parallel or identical lines) there is no unique solution. In this lesson the pairs are always independent, so a single price pair exists.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.
Developing	Carries out the method with prompts; can explain part of the reasoning.

Secure	Works through independently and explains why each step is true.
Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Supporting Adult Guidance v0.1.